Water Resources Research

RESEARCH ARTICLE

10.1029/2024WR038024

Key Points:

- Integral plume models are heavily parameterized, and their results are strongly sensitive to the plume model assumptions and parameters
- The role of bubble-plumes in the interbasin transport is determined by the vertical structure relative to the sill, and vertical flows
- The interbasin transport of injected oxygen cannot be explained without a combination of meteorological and plume forcing

Supporting Information:

Supporting Information may be found in the online version of this article.

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Citation:

Toledo, J. C., Rueda, F. J., Little, J. C., & Ramón, C. L. (2025). Wind-induced versus plume-induced inter-basin exchange—Resolving causal influences in plume-lake modeling. *Water Resources Research*, *61*, e2024WR038024. https://doi.org/10.1029/2024WR038024

Received 20 MAY 2024
Accepted 18 JUL 2025

Wind-Induced Versus Plume-Induced Inter-Basin Exchange—Resolving Causal Influences in Plume-Lake Modeling

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Abstract Bubble-plumes are commonly used in lake restoration to alleviate problems associated with hypolimnetic hypoxia. In lakes with multiple basins separated by sills, bubble-plumes have been used locally to boost oxygen levels in individual basins. However, they have the potential to affect inter-basin exchange leading to unexpected results. Our goal is to assess the relative importance of natural versus plume forcing as drivers of exchange. This is critical to evaluate the field-scale performance of bubble-plume systems. We hypothesize that the contribution of bubble-plumes as drivers of inter-basin oxygen transport depends on the depths of detrainment and maximum plume rise relative to sill level (plume geometry). To test this hypothesis, 1D integral bubble-plume models coupled to a 3D-hydrodynamic model are applied to simulate the performance of an oxygenation system installed in 1990 in the north basin of Amisk Lake, Canada. Sources of uncertainty in bubble-plume modeling, associated with model assumptions and parameter values (structural and parametric uncertainty), are systematically analyzed, and their effect on plume-structure and inter-basin exchange predictions are quantified. The effects of plume forcing on exchange rates and patterns are only significant as the equilibrium depth rises above the sill. For the prevailing conditions in the study case and the most widely accepted plume model, this occurs with low probability. More plausibly (for most parameter combinations), the plume injects oxygen below the sill, the oxygenated water being transported by internal waves between basins. Solid conclusions on the dominant drivers of large-scale transport arise in attribution studies accounting for model uncertainty.

Plain Language Summary Bubble-plumes, commonly used in lake restoration to avoid hypoxic conditions in eutrophic systems, are known to induce mixing in the hypolimnion, but they also have the potential to reshape larger-scale circulation patterns. Amisk Lake, Canada, is used as a case example where an oxygenation system was deployed in 1990 in one of the two basins caused the injected oxygen to move through the shallow sill into the second basin. Two contrasting interpretations have been proposed regarding the effects of bubble-plumes on exchange for that case, differing on the role given to the bubble plume as driver of interbasin exchange. Those interpretations lie in predictions from two different integral bubble-plume models. Here we systematically analyze the effect that specific assumptions taken in the formulation of integral bubble-plume models (structural uncertainty) and the particular choice of parameter values (parametric uncertainty) can have on the predicted behavior of the plume (level of plume rise, induced recirculation rate and the level at which the oxygenated water is injected). Coupled with a 3D lake hydrodynamic model we quantify the effect of the plume on exchange rates, depending on its behavior. Model configurations and parameters leading to significant changes in interbasin exchange are the least probable.

1. Introduction

Bubble-plumes are commonly used to manage water quality in lakes and reservoirs, with perhaps the most frequent purpose being to alleviate problems associated with hypolimnetic hypoxia in stratified systems (Mobley et al., 2019; Singleton & Little, 2006). Fine oxygen bubbles in these systems are injected through porous diffusers deployed on the lake bottom, creating a gas/water mixture that rises and gains momentum due to positive buoyancy. As it rises, oxygen is transferred from the bubbles into the water and the buoyant mixture in the bubble-plume entrains the surrounding water, increasing the plume's water flow rate and cross-sectional area, while decreasing its vertical momentum. The plume rises against the vertical density gradient until it reaches a depth (the depth of maximum plume rise, d_{mpr}), where the plume's momentum approaches zero. At this depth, the plume water becomes negatively buoyant and sinks back to a depth (the equilibrium depth, d_{eq}) where its density matches the ambient density (McGinnis et al., 2004). Upon reaching d_{eq}, it is assumed that the oxygenated water

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in the plume horizontally intrudes into the far field (Singleton & Little, 2006). As a result of the entrainment and detrainment fluxes in and out of the plume, large-scale mixing is induced. This helps spread the injected oxygen in the hypolimnion. But at the same time, it changes the far-field stratification, and, also the vertical structure of the plume, as described above. A feedback loop is therefore established in which stratification and plume behavior are continuously changing (McGinnis et al., 2004) that must be considered in the design and operation of bubble-plume systems.

Bubble-plume hypolimnetic oxygenation systems can be expensive to implement and operate, and, hence, they need to be carefully designed, and their performance closely monitored and evaluated. In some cases, they might yield unexpected results or even prove ineffective. These cases should be well-documented, studied, and even reanalyzed in light of new tools and concepts as they emerge, since they can provide extremely important lessons for future endeavors. Amisk Lake (Figure 1a), Canada, is an example where a bubble-plume oxygenation system yielded unexpected results. The system was installed in 1990 to boost oxygen levels locally in one of the two basins of the lake, assuming that horizontal oxygen transport would be hindered by the sill separating the two basins. However, dissolved oxygen concentrations increased in both basins. Two mechanisms—plume-induced circulation and wind-induced seiche pumping—had been previously identified (Lawrence et al., 1997) as the most plausible drivers of oxygen transport between the two basins under summer-time stratified conditions (Figures 1a and 1b). Building upon these earlier results, one could infer that inter-basin oxygen transport was largely driven by plume-induced circulation. Those conclusions were based on the interpretation of a spatially sparse experimental data set collected in 1991, using scaling arguments and simplified models of relevant processes, and introducing reasonable assumptions where experimental data was unavailable (Lawrence et al., 1997). More recently, Toledo et al. (2021) used 3D coupled plume-lake modeling tools to reevaluate the conclusions of earlier studies on the performance of the bubble-plume oxygenation system in Amisk Lake. Plume-lake models combine one-dimensional integral bubble-plume models with large-scale three-dimensional hydrodynamic models so that they explicitly account for the interactions between the bubble-plumes and the large-scale transport (Chen et al., 2017; Singleton et al., 2010; Toffolon & Serafini, 2013; Toledo et al., 2021). Based on the analysis and visualization of model simulations, and in contrast with the results of Lawrence et al. (1997) and Toledo et al. (2021) concluded that inter-basin oxygen transport was largely the result of seiche-pumping. The apparent contradiction between the existing interpretations (Lawrence et al. (1997) vs. Toledo et al. (2021)) regarding the contribution of bubble-plumes as drivers of inter-basin oxygen exchange in Amisk Lake has not been resolved.

Given the intrinsic complexity of the hydraulic system, the scarcity of field data, the very own structure of the modeling tools used and their high level of parameterization, the existing interpretations are inherently uncertain. The work of Lawrence et al. (1997) relied on results from a simpler single-plume mixed-fluid model proposed by Schladow (1992) (S92 from here on), which ignores some key processes (such as gas-water mass transfer) explicitly represented in the one-dimensional integral hybrid two-fluid double-plume model of Socolofsky et al. (2008) (SK08 from here on) used by Toledo et al. (2021) (see detailed model descriptions in Section 2 and Figure 1c). It remains unclear whether the structural differences between plume models (structural uncertainty) could account for the divergent interpretations reached by Toledo et al. (2021) and Lawrence et al. (1997) regarding the contribution of bubble-plumes as drivers of inter-basin exchange. It is also unclear whether the selected values for the input model parameters (parametric uncertainty) or the diffuser configuration could affect these conflicting interpretations. Toledo et al. (2021) only used fixed parameter values recommended in SK08. They further assumed fixed operational parameters, and, a plume with circular cross-section, while the diffuser was rectangular in shape, 15 m long and 2 m wide. Assessing model uncertainty becomes crucial in studies aimed at identifying the mechanism/s driving particular features of large-scale transport, a.k.a. attribution studies, especially when multiple potential drivers are at play and field data is limited.

Our goal is to evaluate the contribution of bubble-plumes as drivers of large-scale transport across shallow sills in stratified multi-basin lakes. Amisk Lake is taken as a case study. We hypothesize that the contribution of the plume as a driver of exchange will largely depend on the predicted depth of maximum plume rise, d_{mpr}, and the equilibrium depth, d_{eq}, relative to the depth of the sill, d_{sill} (see Figure 1). Only if the plume rises above the sill will it have the potential to induce horizontal density variations and water movement across the channel through convective readjustment. We rely on the analysis of a series of plume and lake modeling exercises in which uncertainty is explicitly evaluated and accounted for. Our study holds broader significance for several reasons. First, structural differences among existing plume models have not been thoroughly scrutinized and compared using statistical approaches (structural uncertainty). Second, we introduce a detailed methodology that should be

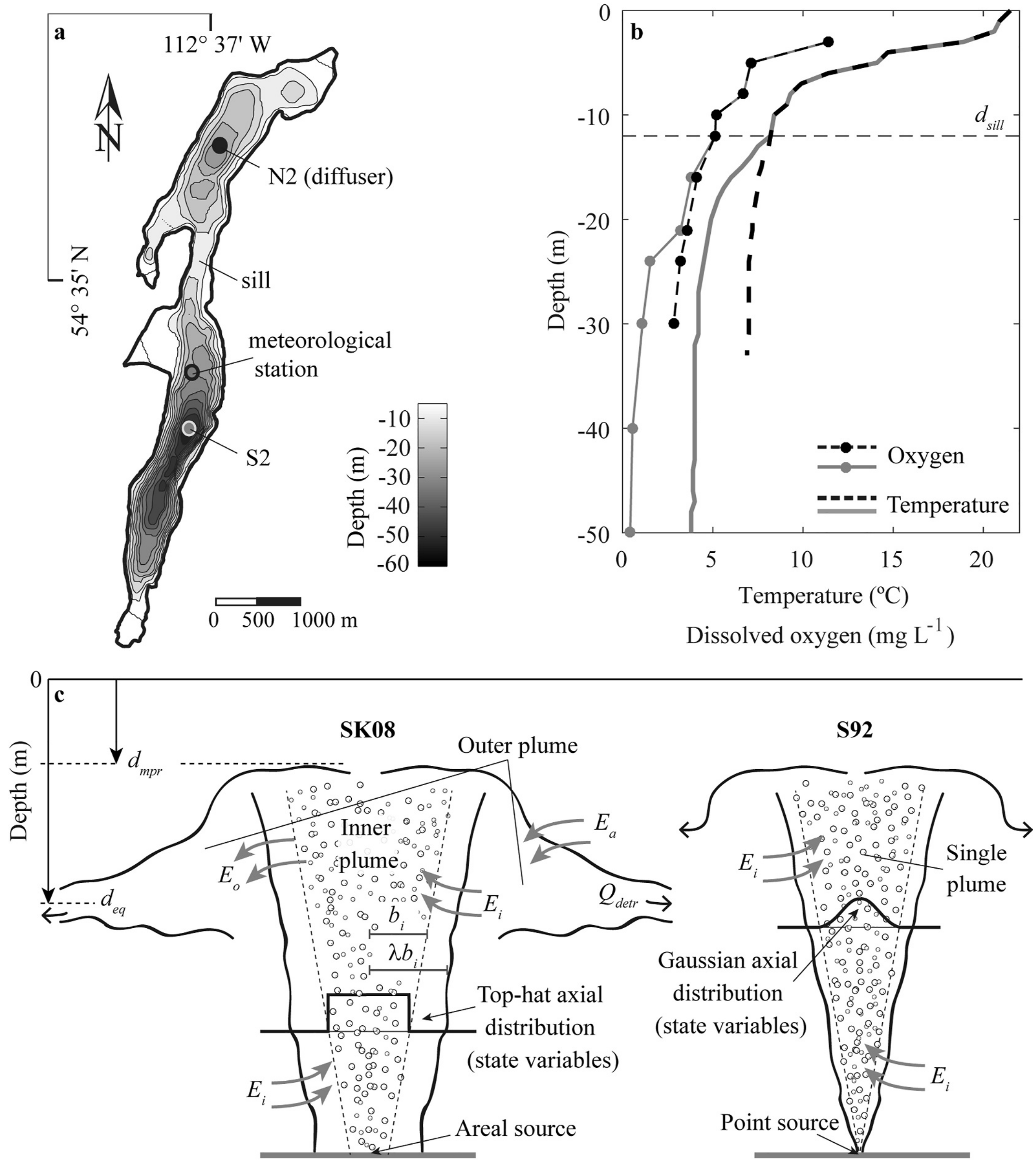

Figure 1. Study site and sketches for the two integral bubble-plume models. (a) Geographic location and bathymetry of Amisk Lake, Canada. Also shown are the location of the meteorological station and measurement sites S2 and N2 in the data set collected in 1991 by Lawrence et al. (1997). (b) Temperature and dissolved oxygen profiles collected on day 181 in 1991, at sites S2 (solid lines) and N2 (dashed lines). (c) Sketches for the double-plume SK08 and single-plume S92 models. Several attributes of the idealized bubble-plumes are identified, including the axial distribution of state variables, the bubble-plume horizontal length scale b_i and the overall plume horizontal length scale λb_i, as well as the entrainment fluxes (E_i, E_o, E_a), the depth of maximum plume rise d_{mpr}, equilibrium depth d_{eq}, and detrainment discharge Q_{detr}.

carried out in plume-lake modeling exercises, given the significant uncertainty surrounding some of the plume-model parameters (parametric uncertainty). This methodology is, in particular, necessary in the reanalysis of previously collected data sets, as some information regarding diffuser configuration and operation may be incomplete.

2. Plume-Model Descriptions

The two integral-plume models used in previous studies of Amisk Lake (S92 and SK08) are presented, and their differences are identified. The single-plume S92 model is earlier and simpler compared to the double-plume SK08 model, which is the most comprehensive integral model for bubble plumes developing under stratified conditions with negligible cross-flows.

2.1. The Single-Plume S92 Model: Starting Point

The S92 model represents the rising motion of water with bubbles, ignoring the subsequent sinking of water that occurs at the depth of maximum plume rise (see sketch in Figure 1c). This model does not account for water-gas transfer, and the gas mass flow rate remains constant throughout the water column. Consequently, the volumetric gas flow varies with depth solely due to isothermal expansion. The model further assumes a Gaussian self-similar axial distribution for both plume velocity and density deficit. It employs a closure based on the formulation of a buoyancy flux conservation equation (see details in Supporting Information S1 from here on). The model equations, on which S92 is based, represent the conservation laws for water-volume flux Q, gas-volume flux q, momentum flux M, and buoyancy flux $\dot{B}$ for a rising bubble-plume. Appendix A (Table A1) provides a summary of the key equations in the S92 model, expressed as functions of the axial coordinate r and the height above the diffuser z.

2.2. Double-Plume SK08 Model: Most Complete Integral Model

The double-plume model proposed by SK08 assumes top-hat axial distributions for all variables and considers two separate sets of equations for the rising bubble-plume, referred to as the inner plume, and the subsequent sinking of dense water, known as the outer plume (see sketch in Figure 1c). Here, we adopt the discrete peeling model configuration proposed by SK08, wherein all the upward flow within the inner plume is assumed to be injected into the outer plume at d_{mpr}, where momentum becomes zero. Appendix A (Table A2) shows a summary of the key equations in the SK08 model, where subscript i, o, and a denote variables pertaining to the rising inner bubble-plume, the outer plume and the ambient water, respectively. In principle, the constants of proportionality, α_i, α_o, and α_a (Table 1), for the different entrainment fluxes (E_{i}, E_{o}, and E_{a} in Figure 1c and Table A2) are independent of each other. However, SK08 proposed using equal values for α_o and α_a, with the entrainment coefficient into the inner plume α_i being half the value assigned to the other entrainment coefficients. For the sake of structural comparison, we assume that all entrainment coefficients share the same value. However, in the analysis of parameter uncertainty, the entrainment coefficients will be treated as independent and random variables (i.e., $\alpha_i \neq \alpha_o \neq \alpha_a$). The buoyant forces resulting from the entrained fluid in the momentum flux equation are calculated by tracking each component that affects water density (i.e., salinity and temperature) and calculating water density and buoyancy forces through an equation of state.

The void fraction used in the calculations of mixed-phase density is estimated from the local volumetric gas flow rate and the depth-varying mass flux of gases, using the ideal gas law. The depth-dependent local mass flux of gases, in turn, is determined using the discrete bubble sub-model proposed by Wüest et al. (1992) (see their Equations 8–10). This model explicitly considers variations in bubble size as they ascend in the water column due to gas-water mass transfer and links the rising speed of bubbles u_s to bubble diameter (see Equations in Table 3b of Wüest et al., 1992). It is important to note that the utilization of the ideal gas law establishes a relationship between the temperature flux equation and the volumetric gas equation. Neither of these effects—namely, changes in bubble slip velocity or the influence of temperature on the volumetric gas flow rate—are accounted for in S92.

The SK08 was recently adapted for linear diffusers (Dissanayake et al., 2021) and is considered one of the most comprehensive integral plume models in the literature for simulating bubble plumes in stratified waters with negligible cross-flows. As evidence of this, the stratified plume module in the Texas A&M Oilspill Calculator (TAMOC, Socolofsky et al., 2015) currently uses the dynamic equations presented and validated in SK08. The TAMOC model is an open-source model utilized by the U.S. National Oceanic and Atmospheric Administration (NOAA) and industry to simulate the fate and transport of gas and oil mixtures from discharges and leaks (Dissanayake et al., 2023; Loranger et al., 2021; Spier et al., 2013).

2.3. Initial Conditions

A continuous point source of bubbles located at the depth of the diffuser was used to initialize the equations in the S92 model. The initial conditions for water centerline velocity and plume radius were determined based on the power series expansion originally proposed by McDougall (1978). It is worth noting that this expansion is only

Table 1
Plume Model Parameters for the Evaluation of Parametric Uncertainty and Range of Values Used in This Study and Reported in the Literature

Parameter	Definition	Range	Literature values	
λ	Ratio of "inner" to overall plume radius (Equations A3, A4, and A7–A11 in Table A1, Equations A15, A19, and A23 in Table A2)	[0.3–1]	0.8–0.9	Milgram (1983)
			0.8	Wüest et al. (1992), Socolofsky et al. (2008)
			0.3	Schladow (1992)
α_i	Entrainment coefficient—into the inner plume—(Equation A6 in Table A1 and Equation A12 in Table A2)	[0.05–0.2]	0.05	Socolofsky et al. (2008)
			0.05–0.2[a]	Wüest et al. (1992)
			0.075–0.102	Fannelop and Sjoen (1980)
α_o	Entrainment coefficient—into the outer plume from inner plume—(Equation A13 in Table A2)	[0.05–0.2]	0.11	Socolofsky et al. (2008)
α_a	Entrainment coefficient—into the outer plume from ambient fluid—(Equation A14 in Table A2)	[0.05–0.2]	0.11	Socolofsky et al. (2008)
γ	Momentum amplification factor (Equation A7 in Table A1, Equations A19 and A20 in Table A2)	[1–2]	1–2	Swan and Moros (1993)
			1.06	George et al. (1977)
			1.1	Socolofsky et al. (2008)
F	Initial densimetric Froude number of the inner plume (Equation A23 in Table A2)	[1.0–2.0]	1.0–2.0[b]	Wüest et al. (1992)
			1.6	Socolofsky et al. (2008)
F_o	Initial densimetric Froude number of the outer plume at d_{mpr} (Equation A24 in Table A2)	[0.05–0.2]	0.1	Socolofsky et al. (2008)
A_0 (m^2)	Initial inner core (inner plume) area (Equations A10 and A11 in Table A1)	[18–72][c]	36	Lawrence et al. (1997)
q_0 ($m^3 s^{-1}$)	Initial gas flow rate (Equations A10 and A11 in Table A1)	50%–200% ref	0.0028[c]	Lawrence et al. (1997)
r_0 (μm)	Initial bubble radius (to initialize bubble slip velocity in SK08, Table A2)	20–1,500	200–1500[c]	Lawrence et al. (1997)
A_r	Aspect ratio (length to width for rectangular plume)	[1:1–1:12]		

[a]Range in previously published sensitivity analysis. [b][0.1–7] to simulate point sources. [c]Field values.

valid near the source ($z_0 \rightarrow 0$) and under negligible stratification conditions. The calculation of the initial plume radius b_0 and velocity at the source z_0 can be found in Table A1. The reduced gravity at the source g'_0 is estimated from the initial plume radius and velocity and the initial gas flow rate, assuming that the density deficit φ is exclusively driven by the gas content in the plume (i.e., $\varphi = 0$). Note that in the initialization procedure for b_0 and z_0 the solution depends on the choice of the parameters α, the ratio of the "inner" and overall plume radius λ, and the initial gas flow rate q_0 (see Table A1), but also on the slip velocity u_s and the distance to the source z_0.

To initialize their inner plume model, SK08 followed the procedure proposed by Wüest et al. (1992) for an areal source. In this approach, a densimetric Froude number F (Table A2) is used to estimate the initial velocity of the inner plume w_0. The inner core of the plume, where bubbles concentrate, is assumed to be circular (circular plume, with radius λb_0) or rectangular (rectangular plume, with an area $\lambda b_0^2 A_r$, where A_r represents the width-to-length aspect ratio) and initially equal to the area occupied by the diffusers. According to Wüest et al. (1992), realistic initial conditions are achieved by setting $F = 1.6$. To initialize the outer plume at d_{mpr}, Socolofsky et al. (2008) defined an outer Froude number F_o (Table A2) at d_{mpr}. They argued that setting $F_o = 0.1$ results in a smooth variation of state variables from the inner to the outer plume. F_o is then used to calculate the initial vertical velocity of the outer plume w_o, which is in turn used to calculate the volumetric flux in the outer plume Q_o ($=Q_i$) at d_{mpr} (Table A2).

3. Materials and Methods

3.1. Sampling Strategy and Experiment Approach

Two ensembles of m_1 ($=10^4$) and m_2 ($=10^5$) 1D bubble-plume simulations, each one conducted with an independent and random realization of the set of uncertain plume-model parameters (details in Sections 3.2 and 3.3), were generated. The parameter space was sampled using a Latin Hypercube Sampling technique (McKay et al., 1979) from independent uniform distributions spanning the ranges previously established for each parameter (Table 1). Ensemble m_1 was used in the structural uncertainty analysis and ensemble m_2 in the SK08 parameter uncertainty analysis. Thus, each change in model structure also includes its associated parameter uncertainty. Each model run in an ensemble consists of the simulation of the plume formed in Amisk Lake, in response to the operational rules for bubble injection and the prevailing conditions in the water column reported for Amisk Lake during the summer of 1991 (Lawrence et al., 1997; Prepas et al., 1997; Toledo et al., 2021, Figure 1b). All models were solved using the second-order adaptive-step solver for stiff ODEs, ode23s in Matlab, based on the Runge-Kutta-Fehlberg method. In the case of the double-plume model, the inner and outer plume equations are solved separately, with convergence achieved (the inner and outer plume reached a stationary solution), after three iterations.

We evaluated the effect of structural and parametric uncertainties on three model outputs—d_{mpr}, d_{eq}, and the entrainment flow rate Q_{entr}—which were analyzed individually. The equilibrium depth d_{eq} was estimated as the depth where the temperature at the end of the plume calculations (at d_{mpr} in the case of a single-plume, or the bottom of the outer plume in the cases of a double-plume model) equals that of the ambient water. The entrainment flow rate Q_{entr} was calculated as the detrainment rate Q_{detr} minus the initial water flow rate. These three output variables were deemed most relevant as they represent various aspects of the impact of plume operation on the large-scale thermal structure of a lake or reservoir. For instance, d_{mpr} defines the extent of the water column affected by entrainment/detrainment fluxes, where changes in the thermal structure are expected to occur. On the other hand, d_{eq} indicates where the most significant changes (detrainment) are concentrated. Additionally, the magnitude of the detrainment fluxes, which equals the sum of the entrainment fluxes from the environment into the plume, provides insight into the rate at which heat is redistributed below d_{mpr}. In a confined basin, the effect of plumes on the thermal structure (expressed as an effective diffusivity) will depend on the magnitude of Q_{detr}, together with the volume of the basin below d_{mpr}.

3.2. Structural Uncertainty

To identify and assess the primary sources and magnitude of structural uncertainty in bubble-plume modeling, we simulated the bubble-plume diffuser operating in Amisk Lake using alternative plume-model structures. Our starting point in the range of considered modeling structures is the S92 model, which was used by Lawrence et al. (1997) in their interpretation of observations in Amisk Lake. To transition from this modeling approach to the double-plume SK08 model, employed by Toledo et al. (2021), several adjustments are necessary. These adjustments involve addressing the following features: (a) the initial source geometry—switching from point source in S92 to an aerial source (circular plume) in SK08; (b) the assumed axial distribution of state variables—transitioning from a Gaussian distribution in S92 to a top-hat distribution in SK08; (c) the inner structure of the plume—S92 being a mixed-fluid model and SK08 a hybrid two-fluid model (details in Text S1 in Supporting Information S1); (d) the model closure used to calculate buoyant forces in the momentum equation due to bubbles and the entrained fluid—based on a buoyancy flux conservation equation in S92, as opposed to conservation equations for active scalars and equations of state in SK08; (e) the calculation of gas expansion—switching from isothermal in S92 to temperature-dependent in SK08; (f) the calculation of the bubble rise velocity—being constant in S92 and dependent on bubble size in SK08; (g) accounting for gas-water mass transfer of gas components (mainly nitrogen and oxygen) in SK08; and (h) explicitly addressing and modeling transport and mixing processes occurring as the plume sinks from the point of maximum plume rise. An additional step (9) is added where we evaluate the shape of the diffuser, switching from a circular to a rectangular plume. The equivalence rules for assessing structural changes (1–2) and the specifics of structural modifications (3–4) are outlined in Text S1 in Supporting Information S1. Note in Text S1 in Supporting Information S1, that the ensemble of parameters used in the Gaussian model simulations is derived from the ensemble used in all other model configurations (based on the assumption of top-hat variable distribution). Only the ratio of length scales, λ_G and λ_T, are independent in the ensemble of parameters used in the Gaussian and top-hat models, where the subscript G and T refer to Gaussian and top-hat, respectively. The structural changes (1–9) were implemented sequentially.

For each model structure under consideration, we generated an ensemble m_1 of 10^4 simulations. Each ensemble involved an independent and random realization of a set of parameters previously identified as plausible sources of uncertainty (Table 1 and see Section 3.3). Model structures were compared in a statistical framework, in which

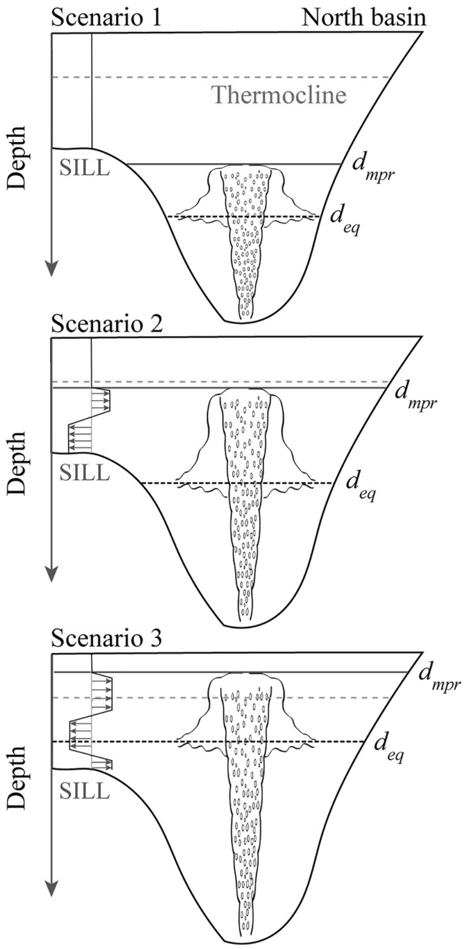

Figure 2. Plume scenarios to analyze the effect of the plume on interbasin exchanges in Amisk Lake. The plume scenarios are defined by the depth of maximum plume rise, d_{mpr}, and the equilibrium depth, d_{eq}, relative to the depth of the sill, d_{sill}. Scenario 1: $d_{eq} < d_{mpr} < d_{sill}$, scenario 2: $d_{eq} < d_{sill} < d_{mpr}$ and scenario 3: $d_{sill} < d_{eq} < d_{mpr}$. The profiles depicted in the sill region show the anticipated average plume-driven effect in the interbasin flow exchange for each scenario.

the ensembles of simulations for each model structure were considered experimental repetitions. We utilized the non-parametric Kolmogorov-Smirnov KS Test (Steel & Torrie, 1992) to compare the output distributions predicted by any two given modeling approaches that differ in a particular structural feature. A structural feature was considered to have a significant impact on d_{eq}, d_{mpr}, or Q_{entr} if the differences between the cumulative distributions in the KS test were statistically different for a confidence level of 95% (p-values < 0.05). The p-values and the k-statistics were further used as summary measures of sensitivity, as in McIntyre et al. (2005).

3.3. Parameter Uncertainty

A total of $n = 11$ parameters were considered as sources of uncertainty in the full SK08 model (see Table 1 for parameters definition and range of plausible values). Note that the term "parameter" is used here broadly, encompassing everything that remains fixed during a numerical experiment, in contrast to "variable" representing everything else that varies in that same experiment. With this interpretation, the initial conditions were also considered as parameters. The range of plausible values assigned to each parameter was intended to include all values reported in the literature or used in earlier sensitivity studies for each parameter. For operational parameters (q_0), the lower and upper limits of the variability intervals were set to 50% and 200% of the unique values reported in the literature for Amisk Lake. Note that the intervals of variability selected for all entrainment constants are equal. When defining the ranges of uncertainty, our aim was to have a parameter space wide enough that realistic model results were not excluded. However, parameter values with no physical sense (e.g., length scale ratios $\lambda > 1$, or momentum amplification factors, $\gamma < 1$) were excluded. For the specific analysis of parameter uncertainty for the SK08 model, an ensemble m_2 of 10^5 simulations was used.

3.4. Uncertainty in Coupled Bubble-Plume Lake Modeling

The full 3D plume-lake model used by Toledo et al. (2021) was employed to simulate a 30-day period in Amisk Lake, incorporating a rectangular or a circular diffuser and considering the realistic forcing conditions in 1991 as reported by Lawrence et al. (1997). The full 3D plume-lake model couples the SK08 model with a 3D shallow-water hydrostatic model (Smith, 2006), where entrainment into the plume and detrainment from the plume are represented as a sink and source of mass in the 3D model, respectively. More details of the 3D model and the coupling can be found in Smith (2006), Singleton et al. (2010), and Toledo et al. (2021).

Our analysis focused on assessing the levels of uncertainty stemming from plume parameters in far-field predictions, particularly interbasin exchange rates and patterns. Three distinct scenarios were identified (Figure 2). In scenario 1 (Figure 2a), where $d_{eq} < d_{mpr} < d_{sill}$ (noting that depth is negative, i.e., d_{eq}, d_{mpr}, and $d_{sill} < 0$), it was anticipated that interbasin exchange would remain unaffected by plume-driven currents. However, in scenarios 2 and 3, where $d_{eq} < d_{sill} < d_{mpr}$ and $d_{sill} < d_{eq} < d_{mpr}$, respectively (Figures 2b and 2c), the dynamics of the plume were expected to influence interbasin exchange. Prior to running the plume-lake model, a factor mapping exercise was conducted with the results of the parameter uncertainty analysis of the SK08 model to identify the region of the input (parameter) space producing plume simulations in the three scenarios. Three sets of O(10^2) simulations each (one for each scenario shown in Figure 2 and plume shape) were conducted with random realizations of the $n = 11$ plume parameters in Table 1. For each realization of the parameter set, three different forcing configurations were tested. Those configurations are referred to as the full, only plume, and only met forcing. In the full-forcing configuration, the model was forced using realistic plume and meteorological conditions as observed in the field. In the only plume forcing configuration, the system was only forced by the plume injecting oxygen and subsequent turbulent mixing. Meteorological forcing, that is, wind forcing and heat flux, was in this case muted.

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In the last configuration (only met forcing), plume-induced mixing was assumed negligible, and the effect of the plume was reduced to injecting dissolved oxygen at the equilibrium depth. In each run, d_{mpr}, d_{eq}, and Q_{detr} were tracked. Time-averaged interbasin horizontal gradients of temperature $\Delta_x T$ above the sill level were calculated to assess the plume-induced vertical mixing in the north basin. The total mass of oxygen injected by the bubble-plume and transported through the channel $\Delta_t O$ was also calculated at the north end of the channel. The average water exchange patterns in each scenario were characterized and compared against the exchange profiles reported by Lawrence et al. (1997).

4. Results

4.1. Structural Uncertainty

In Figure 3 we plotted histograms of d_{mpr}, d_{eq}, and Q_{entr} for the sequential changes in model structures transitioning from S92 to SK08, calculated from the $m_1 = 10^4$ ensemble of simulations conducted with random realizations of the model parameters shown in Table 1. In Table S1 in Supporting Information S1 we have compiled the results of the statistical comparisons of individual structural features on the model results. The features that account for most of the significant differences (from a statistical point of view) between S92 and SK08 in Figure 3 and Table S1 in Supporting Information S1 are associated with the initial conditions (point vs. areal finite source, change 0–1), the inclusion of a bubble size-dependent vertical velocity (change 5–6), the simulation of mass transfer (whether it is neglected or accounted for, change 6–7), the explicit representation of the fall-back motion in the double-plume model (change 7–8), and, the diffuser area shape (circular vs. rectangular diffuser area, change 8–9).

A distribution characterized by most of equilibrium heights above the sill (scenario 3 in Figure 2) is only predicted for point sources (column 0 in Figure 3) and in a reduced number of cases with the rectangular diffuser (column 9 in Figure 3). Incorporating a size-dependent bubble rise velocity for an areal source tends to elevate the depth of the maximum plume rise and the equilibrium depths in the water column (column 6 in Figure 3). This shift is primarily attributed to the size-dependent bubble rise velocities predicted by the discrete bubble sub-model proposed by Wüest et al. (1992) being <0.3 m s^{-1} (the constant slip velocity value used in the S92 model, Table A1). Still, none of the model structures initialized with areal sources (columns 1–9 in Figure 3), even with the wide range of parametric uncertainty used (ranges in Table 1), will fall within scenario 3 ($d_{sill} < d_{eq}$) except for a reduced number of parameter combination (less than 4%) for the rectangular diffuser model (see Section 4.3). The rectangular diffuser predicts deeper d_{mpr} than the circular diffuser (columns 8–9 in Figure 3), being scenario 2 the most likely scenario for the double-plume SK08 model.

4.2. Parameter Uncertainty: Double-Plume Model

Two-dimensional histograms plotting the frequency distributions of Q_{entr} d_{eq} and d_{mpr} for different intervals of the input parameters in the ensemble of $m_2 = 10^5$ simulations conducted with SK08 with a rectangular plume are presented in Figure 4. The momentum amplification factor γ (Figures 4j, 4u, and 4f), the entrainment coefficient α_o (Figures 4b, 4m, and 4x), and the initial Froude number for the outer plume F_o (Figures 4f, 4q, and 4b), all have negligible effects on all the output variables considered, at least within the intervals of variability selected for the analysis. In contrast, the operational parameter initial gas flow rate q_0 has, in the range of variability analyzed, a significant impact on the three output variables studied (Figures 4h, 4s, and 4d). The magnitude of entrainment fluxes, Q_{entr}, is largely determined by parameters associated with the initial conditions at the diffuser, controlling the dimensions of the plume and hence the area of exchange (mainly A_r, λ, and A_0, Figures 4d, 4g, and 4k) and by the ambient-outer entrainment constant (α_a, Figure 4c). The effects of other parameters, such as the initial Froude number F used to initialize the water velocity, or the initial bubble radius r_0 are weaker (Figures 4e and 4i). The magnitude of λ and A_r influence the entrainment flow, as it determines the dimensions of the water flow plume, thereby affecting the area of exchange once the initial bubble area A_0 is fixed. Note that the lower and upper limits of the range of values for λ considered in the analysis differ by almost one order of magnitude (Table 1 and Figure 4). Note also that the largest changes in the simulated entrained flows occur for $\lambda < 0.6$ (Figure 4d), where Q_{entr} varies as $\approx\lambda^{-1}$. The fact that the magnitude of α_a exerts a strong control on Q_{entr} suggests that most of the ambient water is entrained into the outer plume. This occurs because (a) the limited distance over which the inner plume is in direct contact with the ambient water, which is much smaller than the length of the outer plume (see

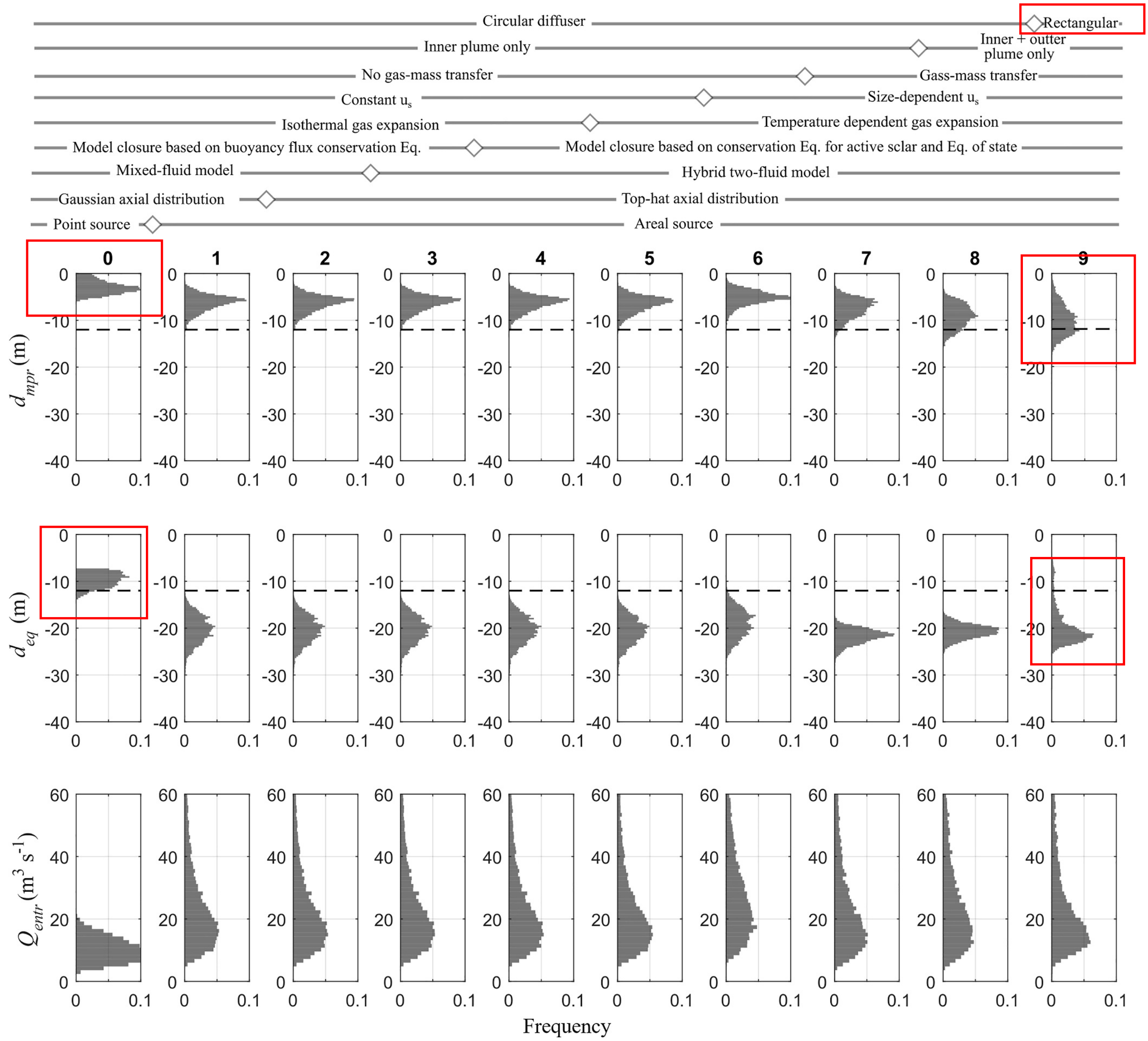

Figure 3. Histograms of the frequency distribution of d_{mpr}, d_{eq}, and Q_{entr} for the different combinations of model structures. The model structural features used on each step are implemented sequentially from the S92 model (column 0) to SK08 model (columns 8–9) and are indicated at the top of the figure.

the differences in d_{mpr} and d_{eq} for the double-plume model in Figure 3) and (b) the radius (hence, the area of contact per unit depth) of the outer plume exceeds that of the inner plume.

In contrast, the coefficient α_i, which parameterizes the exchange between the ambient and the inner-plume, exerts a strong influence on both the vertical development of the plume, d_{mpr} (Figure 4w), and the equilibrium depth, d_{eq} (Figure 4l). The magnitude of the initial Froude number at the diffuser (F), the initial bubble radius r_0 and the scale ratio λ, in this order, also influence the magnitude of d_{mpr}, although their influence is weaker in comparison to α_i and q_0. The sensitivity of d_{eq} to the inner-plume-ambient entrainment coefficient α_i is similar in magnitude to its sensitivity to the initial gas flow rate q_0 (Figure 4dd), aspect ratio A_r (Figure 4gg), initial Froude number F (Figure 4aa) and bubble-plume initial area, A_0 (Figure 4cc).

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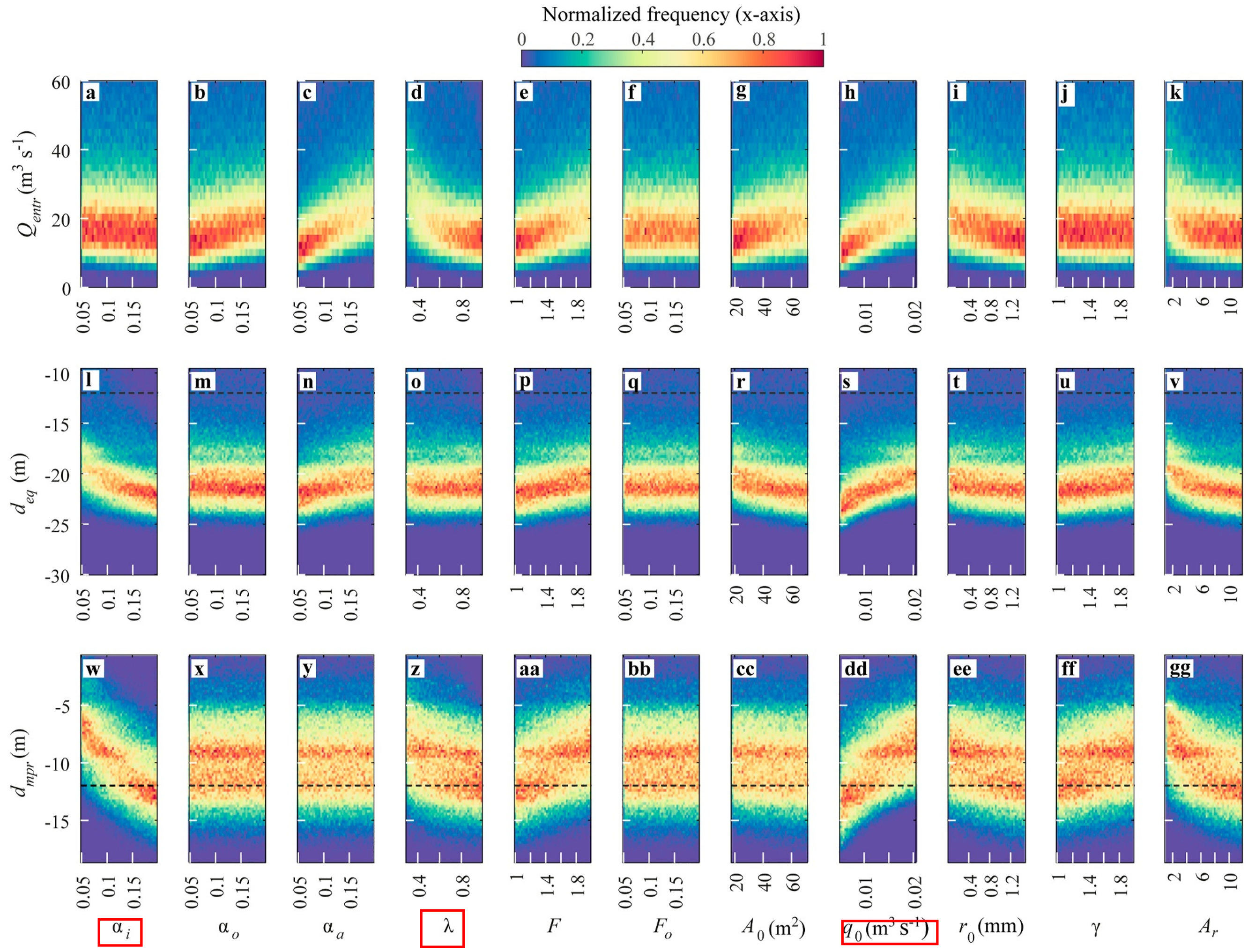

Figure 4. Parametric uncertainty analysis of the double-plume SK08 model. Two-dimensional histograms of the normalized frequency distribution of Q_{entr} (a–k), d_{eq} (l–v), and d_{mpr} (w-gg) for each parameter value evaluated in the parametric uncertainty analysis (with ranges defined in Table 1). Results for a rectangular plume with an ensemble of $m = 10^5$ simulations.

4.3. Plume Scenarios: A Factor Mapping Exercise

We now focus on those eight parameters (α_i, α_a, λ, F, A_0, q_0, r_0, and A_r) to which d_{mpr} and d_{eq} are most sensitive and delineate the regions of the parameter space corresponding to each of the 3 scenarios. The distributions of the parameter values producing simulations in scenarios 1 through 3 are shown in Figure 5 for a rectangular plume. Simulations with $d_{\mathrm{sill}} < d_{\mathrm{eq}}$ (scenario 3) tend to correspond to plumes that entrain large volumes of water from the environment into the outer fall-back plume ($\alpha_a\uparrow$), featuring small aspect ratios ($A_r\downarrow$, therefore smaller areas of exchange for the same initial diffuser area) and low inner plume entrainment coefficients ($\alpha_i\downarrow$) and high initial gas flow rate ($q_0\uparrow$). These last three conditions result in low ambient-inner plume exchange rates and large plume developments with the top of the plume penetrating through the metalimnion into the surface layers. Conversely, simulations with $d_{\mathrm{sill}} > d_{\mathrm{mpr}}$ (scenario 1) correspond to parameter combinations yielding plumes with small vertical developments, characterized by large entrainment rates into the inner plume (larger areas of exchange $A_r\uparrow$ and large inner plume entrainment coefficients $\alpha_i\uparrow$), but low initial velocities ($F\downarrow$) and larger bubble diameters ($r_0\uparrow$ within the range of values reported). This is consistent with the results shown in Figure 4. For the rectangular plume, nearly 25% of the 10^5 simulations corresponded to scenario 1, 71% to scenario 2 and about 4% to scenario 3. Thus, scenario 3 ($d_{\mathrm{sill}} < d_{\mathrm{eq}}$) was the least likely under the stratification conditions existing during the study period in Amisk Lake.

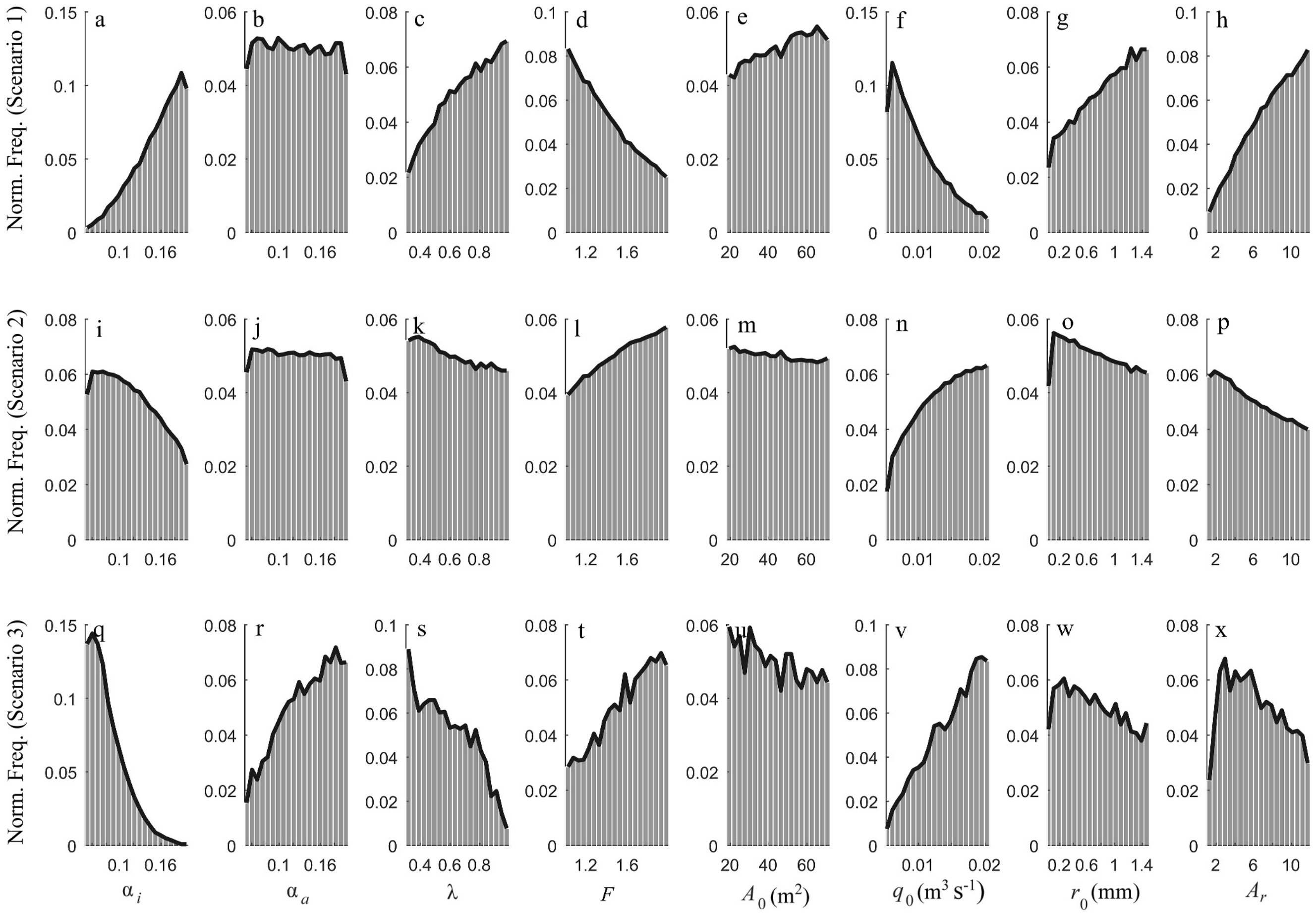

Figure 5. Normalized frequency distribution of the parameters to which d_{mpr} and d_{eq} are most sensitive in the different plume scenarios. Parameter distributions yielding rectangular plumes in scenarios 1 (a–h), 2 (i–p) and 3 (q–x).

4.4. Interbasin Exchange Rates and Patterns: Uncertainty Stemming From Plume Parameters in Far-Field Predictions

During the 30-day period covered by the full-3D plume-lake model simulations, both d_{mpr} and d_{eq} tended to become shallower across all scenarios. Consequently, simulations initially categorized as scenario 1 ($d_{\mathrm{sill}} > d_{\mathrm{mpr}}$) transitioned to scenario 2 ($d_{\mathrm{sill}} < d_{\mathrm{mpr}}$) as the simulation progressed. Transitions from scenario 2 ($d_{\mathrm{eq}} < d_{\mathrm{sill}} < d_{\mathrm{mpr}}$) to scenario 3 ($d_{\mathrm{sill}} < d_{\mathrm{eq}}$) were also observed. Hence, the simulations were reclassified based on the average value of d_{mpr} and d_{eq} throughout the 30 days simulation period. For example, although we initially did not identify any simulations categorized as scenario 3 in the double-plume model with a circular diffuser (column 8 in Figure 3), few simulations initially categorized in scenario 2 showed 30-day-averaged d_{eq} values above d_{sill}, and were recategorized as scenario 3.

In Figure 6 we present the average modeled water exchange patterns in the simulations utilizing only plume forcing for scenarios 2 and 3 (recall that scenario 1 has no plume-induced exchange pattern). In scenario 2 (Figure 6a), the average water flow through the channel exhibits a vertical mode V1 exchange pattern, characterized by a single unique node, separating southward flows near the bottom from northward currents above, developed in response to plumes detraining water below the sill. Conversely, a two-node vertical mode V2 exchange pattern emerged in scenario 3 (Figure 6b), with water flowing from the north to the south basin near the equilibrium depth (d_{eq}) and from the south to the north basin above and below it. This pattern closely resembles the long-term average circulation observed at the sill by Lawrence et al. (1997) using an acoustic Doppler current profiler. Furthermore, the magnitude of the interbasin water exchange rate under only plume forcing in scenario 3 (note the difference of scales used in Figures 6a and 6b) is comparable to that reported by Lawrence et al. (1997), where maximum average southward water exchanges of $\approx -0.25\ \mathrm{m^3\ s^{-1}}$ occurred between depths of 5–8 m.

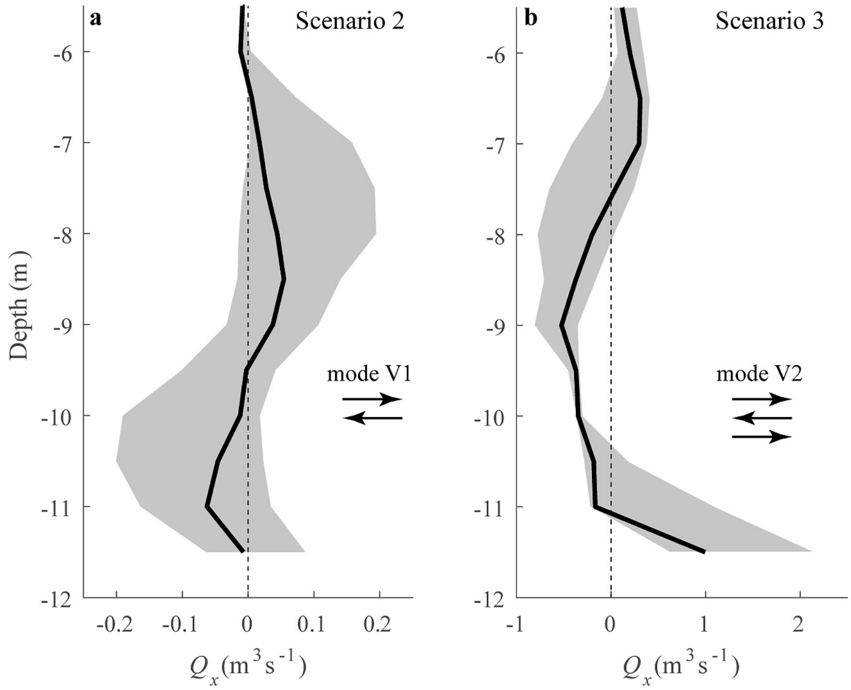

Figure 6. Average interbasin exchange water flow rate through the channel Q_x for scenario 2 (a) and scenario 3 (b) under only plume forcing. The gray shaded areas represent the 25th and 75th percentiles of the range for all simulations in each scenario. Note that the x-axis has a different scale for each scenario.

While these findings could potentially support the conclusion that the observed exchange patterns are driven by plume operation, it is important to note that the likelihood of plume forcing being the main driver of exchange is low. This is because scenario 3 occurs only for extreme parameter values, including very small areal diffuser sources (see factor mapping exercise in Figure 5). Moreover, as argued by Toledo et al. (2021), other processes such as internal-wave pumping could also lead to similar mode V2 exchange patterns when averaged over time through several cycles. These alternative processes could mask or even dominate the effects of the plume in driving changes in stratification and interbasin exchange.

The mass of injected oxygen exchanged between the north and south basins during the 30-day period $\Delta_t O$ is plotted in Figure 7a as a function of d_{eq}. For a given d_{eq}, the estimated magnitude of $\Delta_t O$ is larger in the simulations with full-forcing (filled circles in Figure 7a) compared to other forcing configurations (+ and open circles in Figure 7a). Additionally, $\Delta_t O$ generally increases as d_{eq} rises in the water column across all scenarios and forcing configurations. For $d_{eq} << d_{sill}$, $\Delta_t O$ remains nearly constant regardless of the magnitude of d_{eq} in the simulations with only plume (open circles in Figure 7a) and only met forcing (+ symbol in Figure 7a). It is almost null with only plume forcing, but significantly larger ($\Delta_t O \approx 1.6$ tons) under only met forcing. Interbasin transport in this latter case mainly occurs due to wind-driven internal waves causing oxygen-rich layers (at d_{eq}) to rise above the sill level during the strongest wind events, a mechanism known as internal wave pumping (Van Senden & Imboden, 1989; Wang et al., 2007), which is not present during only plume forcing. The sensitivity of $\Delta_t O$ to changes in d_{eq} varies significantly depending on the position of d_{eq} relative to d_{sill} under all forcing conditions. Small changes in d_{eq} lead to larger changes in $\Delta_t O$ in scenario 3 compared to scenario 2. As suggested in our factor mapping exercise (Section 4.3), bubble-plumes in scenario 3 exhibit large vertical developments with the top of the plume penetrating through the metalimnion into the surface layers and entraining large volumes of water into the outer falling-back rings ($\alpha_a\uparrow$). This leads to large detrainment rates ($Q_{detr}\uparrow$) of oxygenated water at $d_{eq} > d_{sill}$ (larger symbol sizes in Figure 7a). As a result, the plume introduces significant disturbances in the far-field above the sill level, causing large temperature gradients along the channel (see $\Delta_x T$ color distribution, in Figure 7a), accelerating exchange flows, and increasing interbasin oxygen transport.

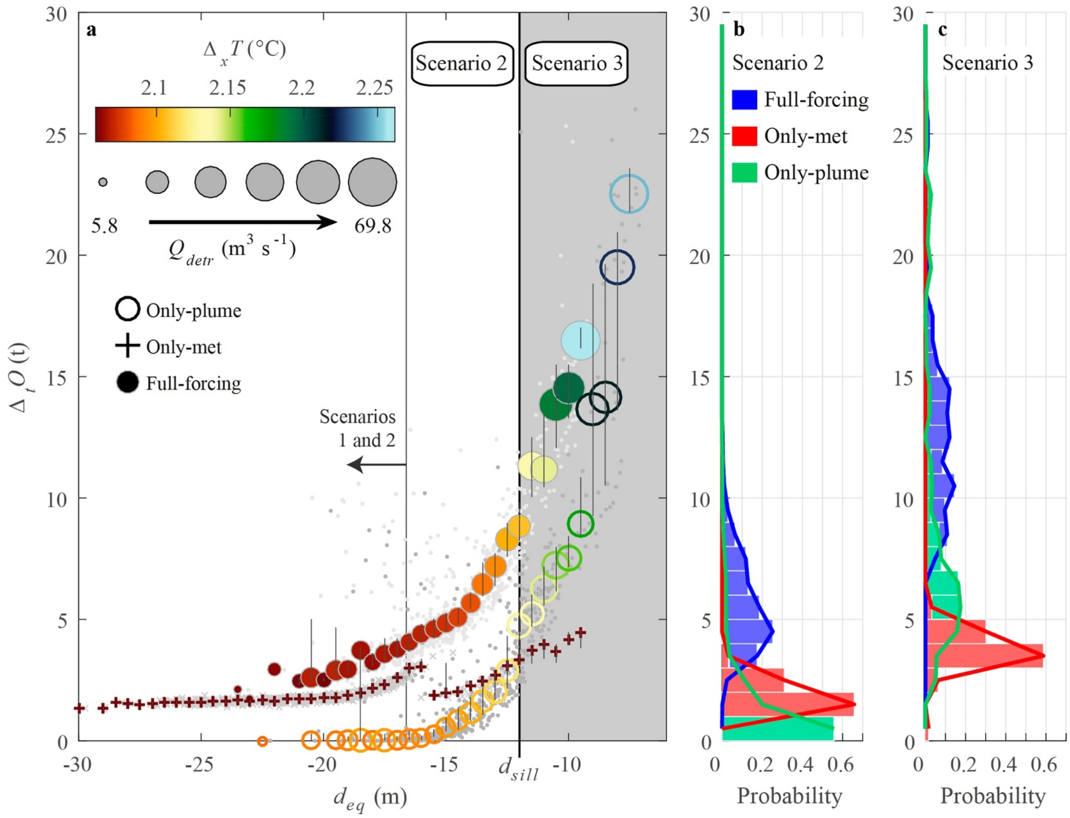

Figure 7. Interbasin exchange of injected oxygen. (a) Mass of injected oxygen $\Delta_t O$ transported through the channel toward the south basin after a 30-day simulation period versus equilibrium depth d_{eq} for the 3D plume-lake simulations using different forcings. Filled circles, open circles and crosses indicate the median total injected-oxygen mass transport for all simulations with full-forcing, only plume forcing and only met forcing, respectively, that reproduced d_{eq} within a given 0.5 m interval. The vertical gray lines on top of each symbol indicate the interquartile range. The color of the symbol represents the average horizontal temperature gradient $\Delta_x T$ between the two basins above the sill level. For the full- and only plume forcings, symbol size represents the average detrainment discharge Q_{detr} for each d_{eq} interval and forcing. Light-gray dots, dark-gray dots and gray *x* symbols in the background show the results for all runs with full forcing, only plume forcing and only met forcing, respectively. Results combine ensembles conducted with a circular and a rectangular plume. The shaded gray region shows simulations with equilibrium depths above the sill (scenario 3). (b, c) Probability histograms of the total injected-oxygen mass transferred to the south basin after a 30-day simulation period for each forcing combination in (b) scenario 2 and (c) scenario 3.

In general, the estimated values of interbasin oxygen transport $\Delta_t O$ with only met forcing depend on d_{eq} and d_{mpr} ($\Delta_t O_{scenario1} < \Delta_t O_{scenario2} < \Delta_t O_{scenario3}$), but they are weaker compared to those calculated with full forcing (see histograms in Figures 7b and 7c for scenarios 2–3, respectively, and Table 2). With only plume forcing (i.e., muting meteorological forcing), the distribution of $\Delta_t O$ is even weaker and skewed toward zero in scenario 1. Only in the most extreme cases of oxygen transport in scenario 3, plume forcing itself could drive similar mass transport compared to those simulated with full-forcing (maximum values of $\Delta_t O = 26.01$ tons with only plume-forcing vs. $\Delta_t O = 25.32$ tons for full forcing in Table 2). In scenario 2, the average magnitude of $\Delta_t O$ with only plume forcing is 65% lower (1.89 tons, Table 2) compared to $\Delta_t O$ under full-forcing conditions (5.59 tons, Table 2). With only met forcing (without plume mixing) it is 75% lower (1.45 tons, Table 2). In scenario 3, the average magnitude of $\Delta_t O$ decreases significantly compared to full-forcing when only met (70% reduction) or only plume (35% reduction) forcing are considered (Table 2). These results suggest that meteorological and plume forcing act synergistically creating interbasin oxygen exchange. A two-way ANOVA test (details in Table S2 in Supporting Information S1) confirmed statistically significant interactions between plume-forcing and wind-forcing on the total mass of injected oxygen that was exchanged between the north and south basins, with higher interaction occurring in scenario 2 than in scenario 3.

Table 2
Mass of the Injected Oxygen in Tons Transported Toward the South Basin After a 30-Day Simulation Period

	Scenario 1			Scenario 2			Scenario 3		
Forcing combination	Mean	SD	Max	Mean	SD	Max	Mean	SD	Max
Only met forcing	1.62	0.09	1.91	1.98	0.40	3.61	3.74	0.74	5.35
Only plume forcing	0.00	0.00	0.00	1.45	1.89	11.27	8.25	5.42	26.01
Full forcing	2.80	0.38	4.04	5.59	1.84	12.81	12.42	3.40	25.32

Note. Average (Mean), standard deviation (SD) and maximum value (Max) for scenario 1, scenario 2 and scenario 3 for the three forcing combinations.

5. Discussion

As hypothesized, the contribution of the plume as a driver of exchange in stratified multi-basin lakes largely depends on the depth of maximum plume rise, d_{mpr}, and the equilibrium depth, d_{eq}, relative to the depth of the sill, d_{sill} (see Figure 2). For inter-basin oxygen exchange to occur, oxygenated water must rise above the sill level. If plume detrainment occurs above the sill level (i.e., $d_{eq} > d_{sill}$), inter-basin oxygen exchange is primarily driven by plume-induced circulation. This exchange occurs continuously and is significant in magnitude (as shown in Figure 7a). However, for plumes that detrain oxygenated water below the sill level, inter-basin oxygen exchange is episodic and largely driven by internal wave pumping. Its magnitude will vary depending on (a) the distance between the detrainment level and the sill, and (b) the amplitude of wind-driven internal waves. Plumes with limited vertical development (i.e., those with a maximum plume rise d_{mpr} below the sill level) are the least likely to drive exchange.

Through sensitivity analysis, we identified the structural features and model parameters that were more determinant in modifying the depth of the maximum plume rise d_{mpr}, the equilibrium depth d_{eq} and the intrusion flow rates in integral 1D plume models. Most of the previous research on plume-lake coupling has evaluated the response of lake stratification to bubble-plume operations using fixed plume-model structures and parameters derived from the literature (i.e., S. Chen et al., 2017, 2018; Toledo et al., 2021). However, these studies have not considered all potential uncertainties inherent to plume modeling, whether structural or parametric. Wüest et al. (1992) and Singleton et al. (2007) conducted a sensitivity analysis in their work, but they assumed fixed environmental conditions, as their work was focused on bubble-plume modeling. Singleton et al. (2010) conducted a sensitivity analysis of their coupled lake(3D)-plume model's temperature and oxygen profile predictions during an 11-day simulation of diffuser operation in Spring Hollow Reservoir. They perturbed only the operational parameters (gas flow rate and diffuser length) and the initial bubble radius in their linear single-plume model. S. Chen et al. (2017, 2018) incorporated a water-jet model into the coupled lake(3D)-plume model developed by Singleton et al. (2010) to assess the impact of operational scenarios involving a bubble-plume epilimnetic mixing system and a side-stream supersaturation hypolimnetic oxygenation system on the thermal structure, dissolved oxygen content, and solute transport within Falling Creek Reservoir. The same parameters used for the linear bubble-plume in Singleton et al. (2007) were employed, without consideration of parametric uncertainties. Furthermore, Singleton et al. (2010) and S. Chen et al. (2017, 2018) only evaluated a single-plume model structure, whereas Toledo et al. (2021) only assessed a double-plume model structure. To the best of the authors' knowledge, no direct comparison between alternative models has been conducted thus far, except for the work of Dissanayake et al. (2021) who compared predictions of a linear single-plume model and a linear double-plume model during the thermal stratification in Spring Hollow Reservoir and showed that the double-plume model better reproduced the intrusion depth and flow rate.

Our parametric analysis (Figure 4) showed that the vertical development of the plume, d_{mpr}, is largely controlled by the magnitude of the ambient-inner-plume entrainment coefficient α_i, initial gas flow rate q_0 and aspect ratio A_r. The magnitude of the entrainment fluxes, Q_{entr}, is largely determined by parameters describing the plume geometry near the source (mainly A_r, λ, and the initial plume area A_0) and by the ambient-outer-plume entrainment constant (α_a). The equilibrium depth d_{eq} is, in addition, sensitive to α_i, q_0, A_r, F, and A_0. Our results regarding the sensitivity of d_{eq} and d_{mpr} are partly consistent with earlier sensitivity studies. Wüest et al. (1992) (from here on W92), for example, analyzed the sensitivity of their single-plume model to four of the parameters tested here (r_0, q_0, A_0, and α_i), and, found that q_0 and r_0 had the largest influence on the model results (see their Figure 7, and

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discussion in their Section 3.2). These results contrast with those reported here, where q_0 and r_0 are shown to have some effect (especially q_0) but are limited in comparison to other parameters. The differing results arise because of the different analysis tools and ranges used. W92 used a local sensitivity approach (not global as done here) in which each parameter was perturbed at a time from a fixed set of parameter values taken as reference. Moreover, the ranges of variation used in W92 were larger than those used here which, in this study, were defined based on reasonable assumptions taken from observations. For example, q_0 was allowed to vary two orders of magnitude from O(10^{-3}) to O(10^{-1}) $m^3\ s^{-1}$ in W92. This range is larger than ours, which spans values that did not even differ by one order of magnitude, being in the lowest of the range of values used in W92. The intervals of variation for r_0 included values differing by almost three orders of magnitude from O(10^{-4}) to O(10^{-1}) m in W92. Here, they only vary from O(10^{-5}) to O(10^{-3}) m. The different tools and ranges used can be partly justified by the different goals of the exercises (this and W92). W92 intended to exhibit general trends in their proposed model. Here, in turn, we are aiming at identifying "reasonable" intervals of variability in the model outputs, based on reasonable estimates of parameter uncertainty ranges. In either case, the trends in d_{eq} and d_{mpr}, whether they increase or decrease in response to increases in model parameters, appear to be consistent here and in W92. For example, d_{eq} and d_{mpr} decrease as bubble diameter r_0 increases, for $r_0 < O(10^{-3})$ m, however d_{mpr} is more sensitive than d_{eq}. They are decreasing functions of α_i and A_0 but increasing in response to F. Note though that the model in W92 is a single-plume model, hence, there is a unique entrainment coefficient.

Our structural analysis (Figure 3) indicated that the conditions prevailing at the diffuser, the explicit representation of mass transfer across the gas-liquid interface, and the fallback motion of water at the top of the rising plume are the three structural features that primarily account for the differences given by the two analyzed bubble-plume models. Of these three, the initial condition (point vs. areal finite) was the only structural assumption that, on average, could lead to a change of plume scenario in Amisk Lake. Only for point sources, as assumed in Lawrence et al. (1997), we were able to simulate plumes that, on average, aligned with scenario 3. The choice of initial plume area (or diffuser area) significantly impacts the solution of SK08 in terms of d_{eq} and d_{mpr}. Moreover, as the diffuser area was decreased from the reported values to near zero, d_{eq} and d_{mpr} predicted by SK08 approached the solutions in S92, with both d_{eq} and d_{mpr} becoming shallower (Figure S2 in Supporting Information S1). The sensitivity of plume structure (and, hence, the predicted scenario) to initial conditions may vary depending on the ratio of the diffuser diameter to the thickness of hypolimnion H_{hip}. For example, by extending the hypolimnion of the north basin of Amisk Lake from 34 m ($H_{hip} \approx 24$ m) to 100 m ($H_{hip} \approx 90$ m), both point and areal sources would negate Scenario 3 (Figure S2 in Supporting Information S1). Plume structure is particularly sensitive to the choice of initial conditions for shallower hypolimnions. In our factor mapping exercise (Figure 5), for example, and for the particular stratification developing in Amisk Lake, a linear diffuser ($A_r \rightarrow \infty$) releasing bubbles with a radius of O(10^{-3}) m would have better contained the injected oxygen in the north basin's hypolimnion (i.e., with the top of the plume below the sill level), thus aligning with the lake managers' intended plan.

By including bubble-plume parametric uncertainty into the coupled lake(3D)-plume model, we statistically assessed the probability that plume-driven circulation alone could drive inter-basin oxygen transport. We also assessed the contribution of anthropogenic (plume-driven) and natural (wind-driven) oxygen transport for different scenarios of d_{mpr} and d_{eq}. Under certain assumptions regarding plume model structure and parameter magnitudes, a modeling exercise could substantiate the hypothesis that a plume could indeed drive significant interbasin oxygen exchange in multi-basin lakes. This was the case, for example, of earlier studies conducted in Amisk Lake (Lawrence et al., 1997). However, when viewed through the lens of an uncertainty analysis, we contend that the probability of those parameter combinations resulting in direct (and substantial) plume-driven oxygen exchange across basins is low. For the particular meteorological and stratification conditions in Amisk Lake during the summer of 1991, scenario 2—where d_{mpr} and d_{eq} are above and below the sill level, respectively—was the most probable. In this scenario, oxygen exchange occurred due to a synergistic interaction between plume and internal-wave forcing. Simulations conducted without plume-driven vertical mixing predicted inter-basin oxygen exchange rates in scenario 2 that were nearly 2.5 times lower than those when all drivers were included. In scenario 3, simulations with plume-driven vertical mixing (and circulation) as the sole driver showed oxygen exchange rates similar in magnitude to those with all forcings and more than 2.5 times higher than those driven by meteorological forcing alone. Hence, scenario 3 is the only scenario in which the interpretation of bubble plumes dominating inter-basin exchange holds true. However, scenario 3 is the least likely under the

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conditions present during the study period in Amisk Lake and only emerged using the highest gas flow rate tested and parameter values at the extreme limits of those considered plausible in the literature.

Large-eddy simulations LES (e.g., B. Chen et al., 2018; Fraga et al., 2016; Xiao et al., 2021; Yang et al., 2016) and direct numerical simulations DNS (e.g., Van Reeuwijk et al., 2016) of bubble and multiphase plumes have been used to resolve plume structures avoiding plume parameterization. However, the applicability of those approaches to simulate at lake-scale, for long simulation periods and with realistic meteorological forcing is unfeasible for DNS and still limited for LES. 1D integral plume models coupled to 3D primitive-equations or RANS (Reynolds-Averaged Navier-Stokes) models reduce computational cost and allow for rapid predictions, long simulations and evaluation of multiple scenarios. Fraga and Stoesser (2016) explored the effects of bubble size, diffuser width, and gas flow rate on three integral plume properties: the vertical evolution of the plume's width, centerline velocity, and mass flux. They used both LES and the integral equations from Rouse et al. (1952) and Bombardelli et al. (2007) to model the evolution of these three properties. Their results showed that LES simulations generally agreed with experimental data and the predictions of the integral equations. However, they reported deviations between LES and the integral equations in the slope of the vertical evolution of the centerline velocity and plume radius, particularly when the plume contained bubbles of multiple sizes. We reproduced the numerical experiments of Fraga and Stoesser (2016) using the SK08 model and demonstrated that, when parametric uncertainty is accounted for, the results of the SK08 model encompass those of LES and experimental data (Figures S3–S5 in Supporting Information S1). Despite the high degree of simplification involved in 1D integral models (i.e., self-similar cross-plume variation for the plume variables and the entrainment hypothesis), double-plume models, as SK08, have been also shown to reasonably reproduce d_{mpr} and d_{eq}, intrusion flow rates and dissolved oxygen distribution in stratified waterbodies with negligible cross-flows (e.g., Dissanayake et al., 2021). Still, there has been a paucity of research examining the impact of oxygen bubble-plumes on the surrounding environment and their coupled evolution (e.g., S. Chen et al., 2018; McGinnis, 2003; Singleton et al., 2010). This is particularly pertinent in confined aquatic ecosystems, such as lakes and reservoirs, where the prevailing environmental conditions that control plume behavior can be altered by the plume itself (i.e., S. Chen et al., 2018; McGinnis, 2003; Singleton et al., 2010). By modifying stratification, bubble-plumes have the potential to generate large-scale motions that could be comparable in magnitude to those caused by other natural forcings. This study focuses on the role of bubble-plumes as drivers of large-scale mixing and circulation in confined water bodies (lakes/reservoirs) and analyzed, for the first time, how (i.e., under which scenarios) they can override the effects of natural forcing in two-basin and multi-basin lakes. The attribution exercise, which aims to identify the dominant drivers of circulation, can only be conducted through modeling, and, given the lack of a priori estimates of relevant parameters, requires a combined uncertainty analysis.

6. Conclusion

The injection of oxygen at the deepest point in the north basin of Amisk Lake directly improved dissolved oxygen concentrations not only in the targeted north basin but also in the south basin. Two mechanisms—plume-induced circulation and wind-induced seiche pumping—have been previously identified as the primary drivers of inter-basin water and oxygen exchanges in the lake. However, there has been controversy over which mechanism predominates. Here, we show that this controversy can be explained by differences in the 1D integral plume models and their inherent structural and parametric assumptions used in previous studies. This study underscores the importance of accounting for both structural and parametric uncertainty in 1D integral plume models when simulating plume–lake interactions and assessing basin-scale effects. We identify the most critical model assumptions (structures) and parameters that govern the vertical development of the plume and its detrainment fluxes. In the specific case of the two-basin Amisk Lake, propagating plume model uncertainty to the lake scale enabled a robust assessment of the installed bubble plume device's contribution to interbasin oxygen exchange. For the particular meteorological and stratification conditions prevailing in Amisk Lake in the summer of 1991, oxygen exchange occurred, most likely, as a result of a synergetic interaction between plume and internal-wave forcing, the contribution of direct-plume induced circulation being minor. Such systematic analysis should be conducted in similar attribution studies, characterized by the scarcity of field data and where the applied modeling tools are highly parameterized and structured.

Appendix A: Primary Model Equations for the S92 and SK08 Bubble-Plume Models

This appendix summarizes the governing equations and initial conditions for the single-plume in the S92 model and the double-plume in the SK08 model (Table A1 and Table A2).

Table A1
The Single-Plume S92 Model

Primary model equations		Parameters and definitions	
Density deficit of the plume relative to the ambient density:			
$\rho_a(z)-\rho(r,z)=\underbrace{\left[\rho_a(z)-\rho_w(r,z)\right]}_{\mathrm{P}(r,z)}+\underbrace{\left[\rho_w(r,z)-\rho(r,z)\right]}_{\Sigma(r,z)}$	(A1)	ρ_a	Ambient water density
		ρ_w	Plume-water density
$\rho(r,z)\approx\rho_w(r,z)(1-\chi(r,z))$	(A2)	ρ	Gas-water mixture density
		$\chi(r, z)$	Local mean gas fraction
Gaussian distribution of the gas-volume fraction, density deficit and velocity deficit:			
$\Sigma(r,z)=S(z)\exp\left(-\dfrac{r^2}{\lambda^2 b(z)^2}\right)$	(A3)	b	Overall plume radius
$\mathrm{P}(r,z)=\varphi(z)\exp\left(-\dfrac{r^2}{\lambda^2 b(z)^2}\right)$	(A4)	λ	Ratio of the “inner” and overall plume radius
$\omega(r,z)=w(z)\cdot\exp\left(-\dfrac{r^2}{b(z)^2}\right)$	(A5)	$S(z)/\rho_r$	Centerline (maximum) gas-volume fraction. ρ_r is a reference density for water
		$\varphi(z)$	Centerline (maximum) water density deficit
		$w(z)$	Centerline (maximum) vertical velocity
Governing equations:			
Water volume conservation:		E_i	Entrainment rate
$\dfrac{d}{dz}Q(z)=\dfrac{d}{dz}\left(w(z)\pi b(z)^2\right)=2\pi\alpha_i b(z)\,w(z)=E_i$	(A6)	α_i	Entrainment constant
		u_s	Bubble slip velocity (=0.3 m s^{-1})
Momentum conservation:		g'	Reduced gravity $g'(z)=g\left\{\frac{S(z)}{\rho_r}+\frac{R(z)}{\rho_r}\right\}$
$\dfrac{d}{dz}\left[\dfrac{1}{2}\pi b(z)^2 w^2(z)\right]=\dfrac{1}{\gamma}g'\cdot\pi b(z)^2\cdot\lambda^2$	(A7)	γ	Momentum amplification factor, which represents the ratio of the total momentum flux to the part of momentum flux not attributable to turbulent fluctuations.
Buoyancy conservation:		N^2	Square of the Brunt-Väisälä frequency characterizing ambient stratification
$\dfrac{d}{dz}\left\{Q(z)\cdot g'(z)\dfrac{\lambda^2}{(1+\lambda^2)}\right\}=-Q(z)\,N^2(z)+\dfrac{q(z)\cdot g}{w(z)+u_s(1+\lambda^2)}\cdot\left\{\dfrac{w(z)}{H-z}+\dfrac{dw(z)}{dz}\left(1-\dfrac{w(z)}{w(z)+u_s(1+\lambda^2)}\right)\right\}$	(A8)		
where		H	Hydrostatic pressure at the source
$q(z)=\dfrac{S(z)}{\rho_r}\dfrac{\pi\lambda^2 b(z)^2}{1+\lambda^2}\left[w(z)+u_s(1+\lambda^2)\right]$	(A9)		

Table A1
Continued

Primary model equations		Parameters and definitions
Initial conditions: point source		
Initial plume radius at the source: $$b_0 = 2\alpha z_0\left[0.6 + 0.01719\left(\frac{z_0}{MH}\right)^{\frac{1}{3}} - 0.002527\left(\frac{z_0}{MH}\right)^{\frac{2}{3}} + \frac{z_0}{H}\left(-0.04609 + 0.000031M^{-1}\right) + \ldots\right] \quad (A10)$$	u_B	$u_B = u_s\,(1+\lambda^2)$
	M	$M = \frac{gq_0(1+\lambda^2)}{4\pi\alpha^2 u_B{}^3 H}$
	q_0	Initial gas flow rate
Initial velocity at the source: $$w_0 = u_B\left(\frac{MH}{z_0}\right)^{\frac{1}{3}}\left[1.609 - 0.3195\left(\frac{z_0}{MH}\right)^{\frac{1}{3}} + 0.06693\left(\frac{z_0}{MH}\right)^{\frac{2}{3}} + \frac{z_0}{H}\left(0.4563 - 0.0105M^{-1}\right) + \ldots\right] \quad (A11)$$		

Table A2
The Double-Plume SK08 Model

Model equations		Parameters and definitions
Entrainment fluxes per unit height (Crounse, 2000):		
Into inner plume: $$E_i = 2\pi b_i \alpha_i w_i \quad (A12)$$	b	Plume length. For a circular plume b = plume radius
Into outer plume from inner plume: $$E_o = -2\pi b_i \alpha_o w_o \quad (A13)$$	α	Entrainment coefficient
Into outer plume from ambient: $$E_a = 2\pi b_o \alpha_a w_o \quad (A14)$$	$w(z)$	Average (top-hat) magnitude of the vertical velocity of the plume
Void fraction:		
	$m_b(z)$	Depth-varying mass flux of gases[b]
$$\chi(z) = \frac{q(z)}{\pi\lambda^2 b(z)^2 (w(z) + u_s(z))} \quad (A15)$$	u_s	Bubble slip velocity (m s^{-1}). u_s is a function of the bubble radius[c]
	$q(z)$	Volumetric gas flow rate
	R	Ideal gas constant
$$q(z) = m_b(z)\cdot R\cdot\theta(z)/p(z) \quad (A16)$$	θ	Temperature
	p	Pressure
Governing equations:		
Water volume conservation:	ρ_i	Density of the mixed-phase in the inner plume
Inner plume: $$\frac{d}{dz}\left(\pi w_i b_i{}^2\right) = E_i - E_o \quad (A17)$$	C	Concentration of the component of interest j (salinity, heat, dissolved oxygen, …)
Outer plume: $$\frac{d}{dz}\left(\pi w_o\left(b_o{}^2 - b_i{}^2\right)\right) = E_a + E_o - E_i \quad (A18)$$	$m_{b,j}$	Local mass flux of constituent j in the dispersed phase

Table A2
Continued

Model equations	Parameters and definitions	
Momentum conservation:		
Inner plume: $$\frac{d}{dz}\left(\pi w_i^2 b_i^2\right) = \frac{\pi g b_i^2}{\gamma \rho_r}\lambda^2\left(\rho_a - \rho_i\right) + E_i w_o - E_o w_i \quad \text{(A19)}$$	$C_{(a,o),j}$	Concentration of constituent j in the fluid entrained into the inner plume, that is, the ambient concentration $C_{a,j}$ where the outer plume is not present (i.e. below the equilibrium depth), or the concentration in the outer plume $C_{o,j}$ between the equilibrium depth and the depth of maximum plume rise
Outer plume: $$\frac{d}{dz}\left(\pi w_o^2\left(b_o^2 - b_i^2\right)\right) = \frac{\pi g\left(b_o^2 - b_i^2\right)\left(\rho_o - \rho_a\right)}{\gamma \rho_r} + E_i w_o - E_o w_i \quad \text{(A20)}$$		
Conservation for temperature (heat) and dissolved constituents (gas or salinity):		
Inner plume: $$\frac{d}{dz}\left(\pi b_i^2 w_i C_{i,j}\right) = E_i C_{(a,o),j} - E_o C_{i,j} - \frac{dm_{b,j}}{dz} \quad \text{(A21)}$$		
Outer plume: $$\frac{d}{dz}\left(\pi\left(b_o^2 - b_i^2\right) w_o C_{o,j}\right) = E_i C_{o,j} - E_o C_{i,j} - E_a C_{a,j} \quad \text{(A22)}$$		
Initial conditions: areal source		
Inner plume: $$F = \frac{w_0}{\sqrt{2\lambda b g\left(\rho_a - \rho\right)\rho^{-1}}} \quad \text{(A23)}$$	b_{o0}	Initial outer plume radius of outer plume at d_{mpr} $b_{o0} = \sqrt{\frac{Q_0}{\pi w_0} + b_{i0}^2}$
Outer plume: $$F_o = \frac{w_0}{\sqrt{\left(b_{o0} - b_{i0}\right) g\left(\rho_a - \rho\right)\rho^{-1}}} \quad \text{(A24)}$$	b_{i0} Q_o	Inner plume radius at d_{mpr} Volumetric flux in the outer plume

Note. Example of model equations for a circular plume. [a]Subscript i identify the variables for the rising inner bubble-plume; subscript o, those of the outer plume; and a, those of the ambient water. [b]Calculated using the discrete bubble sub-model proposed by Wüest et al. (1992). [c]Bubble slip velocity calculated as in Wüest et al. (1992) (Their Table 3b).

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Data Availability Statement

The data presented in the figures, as well as the MATLAB version of the SK08 and S92 plume models, and the MATLAB code used to generate the sample population and conduct the structural and parametric uncertainty analysis, are available for download at Zenodo (https://doi.org/10.5281/zenodo.16659473).

Acknowledgments

The first author received support from U.S. National Science Foundation Grant CBET 1033514 while visiting Texas A&M University. It was during that visit that, under the supervision of Scott Socolofsky, the double-plume model was implemented into the 3D hydrodynamic model. Funding for open access charge: Universidad de Granada/CBUA. We thank the three anonymous reviewers for their thorough comments and suggestions.

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